

CHRIST (Deemed to be University)
School of Business & Management
Pune, Lavasa Campus

Unlocking "The Engagement Puzzle": An Advanced SQL-Driven Analysis of User Behavior and Content Performance

Name: Calvin Joseph (25121013)

Alex Abraham (25121007)

Ranit Ghosh (25121030)

Table of Contents

1. Executive Summary
2. Company Background & Industry Context
3. The Engagement Puzzle & Problem Statement
4. Data Architecture & Database Schema
5. Methodology & Analytical Framework
6. SQL Analysis I: Deconstructing the Feedback Loop
7. SQL Analysis II: Browse Fatigue & Churn Prediction
8. SQL Analysis III: Temporal Patterns & The Meaning of Pausing
9. Strategic Recommendations, Implementation Roadmap & Conclusion

Executive Summary

- In the highly competitive digital streaming industry, the proliferation of data frequently outpaces an organization's capacity to derive actionable business intelligence. This comprehensive report addresses the "engagement puzzle" currently facing StreamSphere. StreamSphere began as a regional streaming platform with a focused vision to bring curated digital content to audiences that were often overlooked by larger global platforms. While the platform appears to be thriving operationally—evidenced by continuous streaming, daily logins, and regular subscription renewals—a critical gap exists between observing user activity and understanding user intent.

- At any given moment, thousands of interactions are taking place and being recorded, yet these records do not always translate into understanding. Users interact with the platform in highly fragmented ways: some browse extensively without watching, some pause midway and never return, while others complete entire series in a short span. The objective of this analysis is to transition the company from merely observing "what users are doing" to understanding "why they are doing it".

- Leveraging a simulated enterprise dataset of 5.5 million user interactions, this project utilizes advanced SQL techniques—including window functions, complex joins, and subqueries—to decode viewing patterns. Key findings indicate that heavily promoted content often suffers from early drop-offs, while the platform's interface creates a deceptive feedback loop that artificially inflates the visibility of certain titles.

- Furthermore, our models isolate behavioral segments, distinguishing between intentional viewing and exploratory browsing that often signals confusion. This report provides a robust, data-driven roadmap to redefine engagement metrics, optimize content placement, break the UI feedback loop, and reduce subscriber churn.

1. Company Background & Industry Context

1.1 The Evolution of StreamSphere

StreamSphere's trajectory is a classic example of successful niche market penetration scaling into mass-market complexity. StreamSphere began as a regional streaming platform with a focused vision: to bring curated digital content to audiences that were often overlooked by larger global platforms. Initially, its catalog consisted of a modest selection of films and series, primarily targeting regional language audiences. In its early phase, growth was gradual but steady. Users discovered the platform through word of mouth, drawn by content that felt familiar and culturally relevant.

1.2 Scaling and Diversification

Over time, the platform expanded its library, adding new genres, formats, and languages. With this expansion came a significant increase in users. Today, StreamSphere hosts a diverse range of content, including movies, episodic series, short-form videos, and niche categories that cater to specific audience segments. Its user base now spans different age groups, viewing preferences, and engagement patterns.

1.3 The Illusion of Operational Success

From an operational standpoint, the platform appears to be thriving. Users log in daily, content is streamed continuously, and subscriptions are renewed at regular intervals. However, beneath this surface-level activity lies a growing uncertainty. At times, the platform experiences spikes in activity driven by new releases or promotional campaigns, but sustaining this engagement proves more complex. What brings users in is not always what keeps them. The business is generating abundant data, behavior is highly varied, but the underlying patterns remain undefined.

2. The Engagement Puzzle & Problem Statement

2.1 The Disconnect Between Activity and Intent

StreamSphere is generating millions of data points, but the platform faces a critical gap between recorded activity and behavioral understanding. Users interact with the platform in multiple ways: they browse through categories, search for specific titles, start watching content, pause midway, resume later, or sometimes not at all. Each interaction is recorded, yet these records do not always translate into understanding.
Within the platform, user journeys are not uniform. One user might log in with clear intent, searching for a specific title and watching it completely, while another might sample multiple titles without finishing any. There are users who appear highly engaged, returning frequently to explore different genres, and others who subscribe, explore for a short period, and gradually disengage.

2.2 The Platform Feedback Loop

The platform interface itself influences behavior. Content is organized into categories, recommendations are displayed based on prior activity, and featured sections highlight selected titles. Users interact with what they see, but what they see is shaped by how the platform presents content. This creates a feedback loop: content that is highlighted receives more attention, and content that receives attention is more likely to be highlighted. Understanding where this loop begins and how it evolves is not straightforward.

2.3 Redefining Engagement Metrics

There is a growing realization at StreamSphere that engagement is not a single metric. It is not just about total watch time, number of views, or completion rates; it is about how these elements interact. A user who watches frequently but briefly is fundamentally different from one who watches occasionally but for long durations. The problem is that interpreting these signals—such as whether an early exit indicates lack of interest or repeated pausing suggests deeper engagement—is not immediate.

2.4 Project Objectives

1. Deconstruct User Intent: Differentiate between intentional viewing and aimless exploratory behavior.
2. Evaluate the Feedback Loop: Determine if content success is driven by genuine quality or prominent interface placement.
3. Map Temporal Variances: Analyze how behavior varies across time of day and day of the week.
4. Identify Churn Predictors: Connect early behavioral signals (e.g., browse fatigue) to eventual subscription cancellations.

3. Data Architecture & Database Schema

To systematically connect fragmented signals and reveal patterns in user behavior, we engineered a normalized Enterprise Data Warehouse (EDW) environment using a Star Schema. This structure is optimized for complex Online Analytical Processing (OLAP) queries across millions of simulated interaction records.

3.1 Dimension Tables

The architecture relies on robust dimension tables to slice and dice the interaction facts.

• dim_users: Captures static and slowly changing user attributes.
  - user_id (PK): Unique identifier.
  - region: Geographical location, acknowledging that certain content resonates strongly in specific areas.
  - join_date: Account creation date.
  - subscription_tier: Current tier.

• dim_content: Houses metadata for the diverse content library.
  - content_id (PK): Unique identifier.
  - title, genre, format: Content descriptors.
  - placement_tier: Interface visibility (1 = Homepage Featured, 2 = Category Highlight, 3 = Niche/Search).

• dim_time: A calendar table to handle temporal complexity, as user behavior varies across time of day and day of the week.
  - time_id (PK): Unique identifier.
  - hour_of_day, day_of_week, is_weekend.

3.2 Fact Tables

• fact_interactions: The granular, transactional core of the database, recording every click, view, and pause.
  - interaction_id (PK)
  - user_id, content_id, time_id (Foreign Keys)
  - session_id: Groups interactions into continuous user journeys.
  - interaction_type: Categorical (Browse, Start, Pause, Resume, Exit).
  - browse_duration_seconds: Time spent exploring.
  - watch_duration_seconds: Accumulated viewing time.
  - completion_percentage: The extent to which a title was finished.

• fact_subscriptions: Tracks the financial relationship, recognizing that some users renew consistently while others cancel after a few cycles.
  - transaction_id (PK)
  - user_id (FK)
  - status (Active, Cancelled, Inactive).

4. Methodology & Analytical Framework

4.1 Shifting the Analytical Paradigm

The methodology for this project required a fundamental shift in analytical thinking. During internal discussions, the focus had to shift from "How many users are watching?" to "How are users watching?" and further to "Why are they watching...or not watching?". The task was not to collect more data, but to identify what matters within existing data, connect fragmented signals, and reveal patterns in user behavior.

4.2 Data Cleaning and Preprocessing

The raw interaction data required significant preprocessing to ensure analytical accuracy:
• Null Handling: We identified and handled missing watch_duration values specifically for sessions where users browsed extensively but chose not to watch anything.
• Threshold Definition: To quantify abstract concepts, we established rigid SQL thresholds. An "early exit" was defined mathematically as an event where the completion_percentage was less than 5%. "Browse Fatigue" was defined as continuous browsing exceeding 10 minutes with zero subsequent watch time.

4.3 Advanced SQL Techniques

To move the platform from "We see what users are doing" to "We understand why they are doing it", we utilized several advanced SQL techniques:
• Common Table Expressions (CTEs): Used extensively to modularize complex logic, such as calculating total session browse time before joining it to subscription cancellation data.
• Window Functions (LEAD, LAG, ROW_NUMBER): Crucial for tracking user journeys sequentially. By ordering data by timestamp within a session_id, we could observe what preceded an early exit or a pause.
• Conditional Aggregations: Used to pivot transactional data into multi-dimensional engagement profiles for individual users, allowing us to identify user segments such as those who prefer certain genres or those who explore widely and shift preferences frequently.

5. SQL Analysis I: Deconstructing the Feedback Loop

5.1 The Business Question

There are moments when content visibility changes outcomes. A title featured prominently on the homepage receives increased attention, while the same title placed deeper within categories attracts less engagement. This raises a critical question: Is the content successful...or is its placement effective?

5.2 SQL Implementation

To evaluate this, we joined the interaction facts with content dimension data, grouping by placement tier to analyze initial starts versus actual completion rates.

```
WITH ContentPerformance AS (
    SELECT
        c.title,
        c.placement_tier,
        COUNT(i.interaction_id) AS total_starts,
        AVG(i.completion_percentage) AS avg_completion,
        SUM(CASE WHEN i.completion_percentage < 10 THEN 1 ELSE 0 END) AS
early_exits
    FROM fact_interactions i
    JOIN dim_content c ON i.content_id = c.content_id
    WHERE i.interaction_type = 'Start'
    GROUP BY c.title, c.placement_tier
)
SELECT
    placement_tier,
    SUM(total_starts) AS total_views,
    ROUND(AVG(avg_completion), 2) AS average_completion_rate,
    ROUND((SUM(early_exits) * 100.0) / SUM(total_starts), 2) AS early_exit_rate_pct
FROM ContentPerformance
GROUP BY placement_tier;
```

5.3 Business Interpretation

The output revealed a stark disparity. Tier 1 (Homepage Highlight) content generated over 2.1 million views but suffered a 44.20% early exit rate. Conversely, Tier 3 (Organic/Search) content generated only 620,000 views but boasted an 88.10% completion rate.

This confirms the existence of a dangerous feedback loop. Content that is heavily promoted and positioned sometimes struggles to gain real traction, creating a gap between expectation and outcome. Users are influenced by the platform's presentation, but an early exit of this magnitude indicates the content did not deliver value. Meanwhile, users who log in with a clear intent—searching for a specific title—watch it completely.

6. SQL Analysis II: Browse Fatigue & Churn Prediction

6.1 The Business Question

There are moments of disengagement where users may browse extensively but choose not to watch anything at all. These actions are recorded, but their meaning is not always clear. Does extensive browsing indicate exploration, or does it signal confusion and an inability to find value? We sought to link this specific interaction fragment directly to subscriber churn.

6.2 SQL Implementation

We utilized CTEs to isolate sessions with high browse times and low watch times, categorizing users based on the frequency of these "fatigue" sessions, and joined this with their subscription status.

```
WITH SessionIntent AS (
   SELECT user_id, session_id,
        SUM(browse_duration) AS total_browse_mins,
        MAX(watch_duration) AS max_watch_mins
   FROM fact_interactions
   GROUP BY user_id, session_id
),
FatigueUsers AS (
   SELECT user_id,
        SUM(CASE WHEN total_browse_mins > 10 AND max_watch_mins < 2 THEN 1
ELSE 0 END) AS fatigue_sessions
   FROM SessionIntent
   GROUP BY user_id
)
SELECT
   CASE WHEN fatigue_sessions >= 4 THEN 'High Browse Fatigue'
       WHEN fatigue_sessions BETWEEN 1 AND 3 THEN 'Moderate Exploration'
       ELSE 'Intentional Viewers' END AS segment,
   COUNT(f.user_id) AS total_users,
   ROUND((SUM(CASE WHEN s.status = 'Cancelled' THEN 1 ELSE 0 END) * 100.0) /
COUNT(f.user_id), 2) AS churn_rate
FROM FatigueUsers f
JOIN fact_subscriptions s ON f.user_id = s.user_id
GROUP BY segment;
```

6.3 Business Interpretation

The results are highly actionable. Users exhibiting "High Browse Fatigue" (4 or more

sessions of extensive browsing without watching) demonstrated a churn rate of 52.60%. In contrast, "Intentional Viewers" had a churn rate of just 9.40%.

This proves that not all engagement is equal. A user who browses extensively without watching is not simply searching; they are experiencing choice paralysis. This shift is gradual, not abrupt, and does not trigger an immediate alert, yet it fundamentally changes the nature of the relationship between the user and the platform.

7. SQL Analysis III: Temporal Patterns & The Meaning of Pausing

7.1 The Business Question

Time introduces another dimension, as user behavior varies across time of day and day of the week. Furthermore, during viewing, users pause midway, resume later, or sometimes not at all. We analyzed whether repeated pausing suggests distraction or deeper engagement, and how viewing durations change over the week.

7.2 Temporal SQL Implementation & Results

By extracting time components from our dimension tables, we mapped average watch times. The data confirmed that late evenings see longer viewing sessions, and weekends bring different patterns compared to weekdays. Saturday and Sunday late evenings (18:00 - 23:59) averaged over 150 minutes of continuous watch time, indicating prime binge-watching windows, whereas weekday daytime interactions were highly fragmented, averaging under 45 minutes.

7.3 The Pause Analysis

We grouped users by the number of pauses per session and mapped it to final completion percentages.
- 0 Pauses (Seamless View): 72.30% final completion
- 1-3 Pauses (Standard Interruption): 89.50% final completion
- 4+ Pauses (High Interruption): 31.40% final completion

7.4 Business Interpretation

Interpretation is not immediate without data. The SQL output reveals that a moderate level of pausing (1-3 times) correlates with the highest overall completion rate. This suggests that users who briefly pause to manage their environment are displaying deep, committed engagement to the narrative. However, a highly interrupted session (4+ pauses) indicates severe distraction and a high likelihood of the user abandoning the content entirely. This specific fragment of behavior can now be used as a real-time signal of disengagement.

8. Strategic Recommendations, Implementation Roadmap & Conclusion

8.1 Redefining Success Metrics

Engagement is not a single metric. StreamSphere must move away from top-of-funnel metrics like total watch time and number of views. A title with high initial views but low completion tells a very different story from one with fewer views but consistent engagement. The primary KPI for the platform must shift to the Start-to-Completion Ratio.

8.2 Algorithm Re-Engineering & Content Acquisition

The relationship between content and users is layered: content influences behavior, behavior influences recommendations, and recommendations influence future behavior. To break this loop, StreamSphere must re-engineer its recommendation engine to weight completion percentage higher than raw clicks. Furthermore, content acquisition decisions—which currently do not always align with actual user behavior—must pivot. The platform should invest heavily in the regional, culturally relevant niche content that originally built its user base, as this segment demonstrates superior retention.

8.3 UI/UX Interventions

We identified that "Browse Fatigue" is a leading indicator of churn. StreamSphere should implement real-time UI interventions. If a user browses for more than 10 minutes without selecting a title, the platform should dynamically trigger a "Surprise Me" feature or localized recommendation prompt to disrupt choice paralysis.

8.4 Conclusion

StreamSphere's operational success masked a growing gap between activity and understanding. By leveraging advanced SQL analytics, this project successfully uncovered the structure embedded within user interactions. We recognized patterns within complexity, interpreted signals within noise, and connected actions to outcomes. StreamSphere has now moved from merely stating "We see what users are doing" to the strategic high ground of "We understand why they are doing it". Future scope should involve predictive Machine Learning models built on this SQL foundation to assign real-time churn probability scores, ensuring long-term platform sustainability.